

# NewsBot Intelligence System: Midterm Report

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## Links

**GitHub repository:** <https://github.com/Tiskalani/ITAI2373-Portfolio>

**Project folder:**

<https://github.com/Tiskalani/ITAI2373-Portfolio/tree/main/ITAI2373-NewsBot-Midterm>

**Video demonstration:**

<https://drive.google.com/file/d/1MXvCwSGUxYjsma8vwFQQ8gi9zK3ylmji/view?usp=sharing>

## Project summary

The NewsBot Intelligence System is an end-to-end natural language processing pipeline that converts a raw news article into structured, decision-ready intelligence. For any article it predicts the category, extracts named entities (people, organizations, places, dates, and money), scores sentiment and emotional tone, and surfaces the linguistic patterns that separate one news beat from another. It integrates all eight course modules: preprocessing, TF-IDF feature extraction, part-of-speech analysis, dependency parsing, sentiment and emotion analysis, multi-class classification, and named entity recognition, behind a single `NewsBot.analyze()` call.

## Results

Trained on the BBC News corpus (about 2,000 articles across five categories), the system reaches 97.5 percent test accuracy with a Linear SVM, with all four compared models above 94 percent. The analysis surfaces consistent, defensible patterns: politics is the most negative beat and entertainment the most positive; business is organization-dominant while sport and entertainment are person-dominant; and tech and business are the most noun-dense. Four bonus extensions add unsupervised topic modeling that rediscovers the categories, a domain-customized named entity recognizer, an interactive Gradio dashboard, and a research experiment showing that grammatical style alone predicts the category at 0.56 accuracy versus a 0.20 random baseline. The deployed classifier is robust: it reports how many words it recognized and returns 'uncertain' for empty, very short, or out-of-scope input instead of guessing, demonstrated live on five real June 2026 news articles that all classify correctly.

## How to run

Open `NewsBot_Intelligence_System.ipynb` in Google Colab, uncomment the install lines in the first code cell, and choose Runtime then Run all (about 3 to 5 minutes on a free CPU). The notebook is self-contained and downloads the dataset automatically, and a copy of the dataset is included in the repository.